

GradeTrack — Manual Testing Guide

Generated from the code as of commit `1b5a5b2` (routes in `internal/handlers/routes.go`, pages in `web/src/App.jsx`, data in `schema.sql` + `seed.sql`). Every expected number below is derived from `seed.sql`; the arithmetic is in the **Footnotes** at the end (marked ¹ ² ...).

Feature coverage: all sessions are present — assessments/terms/categories, bulk entry, monthly register, announcements, CSV import/export, account lifecycle, security hardening, admin dashboard and analytics. Nothing from the feature list is absent. Four API endpoints have no UI and are exercised with curl only (see §6).

Time: full guide ≈ 2 h 45 min (of which ~4 min is waiting for rate-limit windows). **15-minute smoke subset:** `A1 A3 A5 · S1 S3 S6 · T1 T3 T5 TM1 · X1 X3 · AD1 ADM5 · SEC1 SEC4` — run these after any future change.

Legend: $\triangle$ = this test changes data (see §1.4 for reset). **EXPECT** = what you must observe. **IF NOT** = most likely broken layer.

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1. Setup & reset

1.1 Reset the database, start both servers, confirm they're up

All commands from the project root (D:\GradeTrack). Use **Git Bash** for the curl commands in this guide (in PowerShell, `curl` is an alias for `Invoke-WebRequest`; use `curl.exe` there).

```
# 1. Reset DB to seed state (drops + recreates every table)
"/c/tools/mysql/current/bin/mysql.exe" -u root -p < schema.sql
"/c/tools/mysql/current/bin/mysql.exe" -u root -p < seed.sql
# PowerShell equivalent: Get-Content schema.sql | & "mysql"

# 2. Backend (needs .env with DB_DSN and JWT_SECRET; PORT is 8080)
export PATH="/c/Program Files/Go/bin:$PATH"
go run ./cmd/server # → "listening on :8080"

# 3. Frontend, in a second terminal (proxies /api → :8080)
cd web && npm install && npm run dev # → http://localhost:5173/

# 4. Confirm
curl -s localhost:8080/api/health # → {"status": "ok"}
curl -s -o /dev/null -w "%{http_code}\n" localhost:5173/
```

If you later change Go code you must restart `go run`; Vite hot-reloads on its own.

1.2 Seeded accounts

Every password is `password123`. Academic year on both classes is 2026–27.

#	Name	Email	Role	What hangs off it
1	Priya Menon	admin@gradetrack.dev	admin	—
2	Ananya Iyer	teacher1@gradetrack.dev	teacher	owns Algebra II (Mathematics), 5 assessments ¹ , 10 sessions Aug 17–21 & 24–28
3	Rohan Deshmukh	teacher2@gradetrack.dev	teacher	owns Physics 101 (Physics), 5 assessments ¹ , 6 sessions Aug 18–20 & 25–27
4	Aarav Sharma (R1001)	student1@gradetrack.dev	student	Algebra II — avg 78.4% , attendance 80% (8/10) ²
5	Diya Patel (R1002)	student2@gradetrack.dev	student	Algebra II — avg 83.75% , 1 missing, attendance **100%**²

6	Kabir Nair (R1003)	student3@gradetrack.dev	student	Algebra II — 95% / 80%; Physics 101 — 84% / 100% ²
7	Fiona Chen (R1004)	student4@gradetrack.dev	student	Algebra II — 37% / 30% (at risk); Physics — 58% / 66.67% (at risk) ²
8	Ishaan Reddy (R1005)	student5@gradetrack.dev	student	Physics 101 — 94.2% / 83.33% ²
9	Hana Kim (R1006)	student6@gradetrack.dev	student	Physics 101 — 64.25% / 50% (at risk) ²
10	Ian Brooks (R1007)	student7@gradetrack.dev	student, deactivated	Algebra II history: Term 1 only, 63.67% / 80% (4/5) — login is refused

Assessments¹: Algebra II — Assignment 1 (assignment, Term 1, /20), Quiz 1 (quiz, T1, /10), Midterm Exam (exam, T1, /100), Assignment 2 (assignment, T2, /20), Quiz 2 (quiz, T2, /10). Physics 101 — Lab 1 (assignment, T1, /50), Quiz 1 (quiz, T1, /10), Midterm Exam (exam, T1, /100), Lab 2 (assignment, T2, /50), Quiz 2 (quiz, T2, /10).
Announcements: 2 per class (seeded Aug 24/28 for Algebra II, Aug 25/27 for Physics 101).

1.3 Browser prep

- Auth is a JWT in `localStorage` (token + user). Two roles in the same browser profile overwrite each other, so use **one**

normal window + one incognito window (or two Chrome profiles) whenever a test says "in a second browser".

- Hard-reset auth state: DevTools (F12) → Application → Local Storage → `http://localhost:5173` → `delete token` and `user`, then reload. Or use the avatar menu → **Log out**.
- Tokens last 8 h. Login is rate-limited to **5 attempts per minute per email** — if you mistype a password several times, wait a minute (see SEC4).

1.4 Resetting mid-test

Re-run the two `mysql.exe` lines from §1.1 — no server restart is needed. Every $\triangle$ test dirties data; the sections are ordered so read-only tests come first. Sections 2.4, 2.7, 5.2 (partly) and 5.3 mutate. If you abandon a run partway through a $\triangle$ section, reset before resuming, because later EXPECT values assume the exact sequence.

2. Test walkthroughs

2.1 Anonymous (A)

A1 — Login page renders. Steps: open `http://localhost:5173`. EXPECT: redirected to `/login`; centered white card on grey background; blue "G" mark, "GradeTrack", "Sign in to continue"; fields **Email**, **Password**; blue **Sign in** button is disabled until both fields have text; link "No account? Create one". IF NOT: frontend not running / Vite proxy misconfigured.

A2 — Wrong password. Steps: email `student1@gradetrack.dev`, password `wrong`, click Sign in. EXPECT: red banner **invalid email or password**; you stay on `/login`. IF NOT: API down (banner would say "request failed") or auth handler.

A3 — Correct login redirects by role. Steps: sign in as `admin@gradetrack.dev` / `password123`; note the URL; Log out

(avatar menu, top right). Repeat for `teacher1@...` and `student1@...`. EXPECT: `admin` → `/admin` (heading "School overview"); `teacher` → `/teacher` ("Your classes"); `student` → `/student` ("Your classes"). App bar shows a role chip (`admin/teacher/student`) and the user's name with an initials avatar. Log out returns to `/login`. IF NOT: `AuthContext / Home` redirect in `App.jsx`.

A4 — Deactivated account. Steps: sign in as `student7@gradetrack.dev / password123`. EXPECT: red banner **account deactivated**; not signed in. IF NOT: `is_active` check in the login handler.

A5 — Protected routes bounce to login. Steps: logged out, type `/student`, `/teacher`, `/admin`, `/classes/1`, `/account/password` into the address bar one at a time. EXPECT: each redirects to `/login` (no flash of content). `/nonsense` also lands on `/login`. IF NOT: `RequireAuth/RequireRole` guards in `App.jsx`.

A6 — Registration creates a student only. △ (adds one user; harmless) Steps: from `/login` click "Create one". Note the form has **no role picker** and the footer says "Teacher or admin? Ask an administrator for an account." Fill Full name `Test Student`, Email `teststudent@gradetrack.dev`, Roll number `R2001`, Password `testpass123`. Click **Create account**. EXPECT: redirected to `/login` with green banner **Account created — sign in to continue**. Sign in with the new account → `/student` with "Not enrolled in any classes yet" (used again in E4). IF NOT: register handler or the Login page success-banner state. Curl proof that the role field is ignored:

```
curl -s localhost:5173/api/auth/register -H 'Content-Type: application/json' -d '{"name": "Sneaky", "email": "sneaky@gradetrack.dev", "password": "testpass123", "rollNumber": "R2001"}' # → 201 {"id": "...", "name": "Sneaky", "email": "sneaky@gradetrack.dev", "password": "testpass123", "rollNumber": "R2001", "role": "student"}
```

A7 — Duplicate email on register. Steps: register again with email `Student1@GradeTrack.dev` (mixed case), any valid other fields. EXPECT: red banner **already exists** (emails are lower-cased before comparison). IF NOT: `normalizeEmail / unique` key.

2.2 Student (S) — sign in as Kabir Nair,

student3@gradetrack.dev

S1 — Dashboard cards. EXPECT: heading "Your classes"; two coloured cards: **Algebra II** (Mathematics) with stats **95%** average / **80%** attendance, and **Physics 101** (Physics) with **84%** / ****100%****². Each card shows 2026–27. No teacher name or student count on student cards (the API doesn't expose them). IF NOT: `/classes` or `/classes/:id/performance` for students.

S2 — "Across your classes" chart. EXPECT: card titled "Across your classes" with **grouped bars** (Kabir has 2 classes; the radar only appears at 3+): Algebra II avg 95 / attendance 80, Physics 101 avg 84 / attendance 100. Legend "Average %" (blue) and "Attendance %" (orange). Hover a bar → tooltip with both numbers. IF NOT: `SubjectRadar` component.

S3 — Per-class summary blocks. Scroll to the **Algebra II** block. EXPECT: big **95%** (5 marks) and **80%** (8 of 10 sessions); a third stat **Term 2** with green chip ↑ **4.2% vs Term 1**³; the line "Your average: 95% · Class average: not shared with students"; "Marks over time" line chart with 5 points (Jun 5, Jun 19, Jul 10, Aug 7, Aug 21 → 100, 90, 90, 95, 100 %) and a dashed grey line at 95 labelled in the legend "Your average (95%)"; "Attendance calendar" grid of 10 cells — day numbers 17 18 19 20 21 24 25 26 27 28 with **18 and 25 red**, the rest green — and **3 session streak**⁴; sub-line "8 of 10 sessions attended". Physics 101 block: **84%** / **100%**, chip ↑ **3.3% vs Term 1**³, 6 green cells, **6 session streak**. IF NOT: `StudentSummary`, `lib/stats.js` (`termDelta`, `currentStreak`).

S4 — Class page (student view). Steps: click **Open class** on Algebra II. EXPECT: URL `/classes/1`; blue banner "Algebra II — Mathematics · 2026-27" (no teacher name, no student count for students); tabs **Stream | Performance | Marks | Attendance — no Roster tab**. Stream shows two posts by **Ananya Iyer**, newest first: "Reminder: Assignment 3 ..." then "Welcome back! Quiz 2 results ...", each with a relative time (e.g. "4 days ago" on 1 Sep 2026). **No compose box, no Delete buttons**. IF NOT: `ClassPage` TABS map / announcements endpoint.

S5 — Performance tab. EXPECT: card "Your performance" with the same numbers as S3 plus two extra charts: "Term comparison"

(personal) with two term groups — Term 1 overall **93.3**, Term 2 **97.5**; bars per category present (assignments/quizzes/exams)⁵ — and "Your average by category": assignments **97.5**, quizzes **95**, exams **90**, subtitle noting class-wide category averages aren't exposed to students. IF NOT: `/performance categories field / TermComparison`.

S6 — Marks tab. EXPECT: card "Your marks", subtitle "5 shown · 0 missing"; rows newest-first: Quiz 2 (quiz, Term 2, 10 / 10, 100%, 2026-08-21), Assignment 2 (19 / 20, 95%), Midterm Exam (90 / 100, 90%), Quiz 1 (9 / 10, 90%), Assignment 1 (20 / 20, 100%). No Student column, no Edit/Delete, no Export button. Filters: Term → "Term 2" shows 2 rows; Category → "exam" (with Term 2) shows the empty state **No marks match these filters** (used again in E7). Reset both to "All". IF NOT: `/classes/1/marks` student filter.

S7 — Attendance tab. EXPECT: "Your attendance" heatmap identical to S3; "Records — 10 sessions" table newest first with green `present` / red `absent` chips (2026-08-25 and 2026-08-18 absent). IF NOT: `/classes/1/attendance` student filter.

S8 — Account menu / change password. Steps: click the avatar/name → **Change password**. EXPECT: `/account/password`, heading "Change password". Type current `password123`, new short, → inline **Too short — at least 8 characters**; new `password123` → **Must differ from the current password**; new `password1234`, confirm `password1235` → **Passwords do not match**. Click Cancel. (Do **not** actually change it; keep the credentials table valid.) IF NOT: `ChangePassword` page validation.

S9 — Forbidden surfaces (direct URLs). Steps: still as Kabir, type `/teacher` → EXPECT redirect to `/student`. `/admin` → `/student`. Sign in as Aarav (`student1@...`, Algebra only) in an incognito window and open `/classes/2`. EXPECT: red banner **forbidden** with a "Back to your classes" link. `/classes/99` → **class not found**. IF NOT: `RequireClassAccess` middleware.

S10 — Forbidden surfaces (curl). Get a token, then prove the API refuses:

```
TOKEN=$(curl -s localhost:8080/api/auth/login -H 'Content
```

```

-d '{"email":"student1@gradetrack.dev","password":"pass
curl -s -w ' %{{http_code}}\n' localhost:8080/api/classes/2
# → {"error":"forbidden"} 403 (not enrolled in P
curl -s -w ' %{{http_code}}\n' localhost:8080/api/classes/1
# → {"error":"forbidden"} 403 (roster is teacher
curl -s -w ' %{{http_code}}\n' localhost:8080/api/users
# → {"error":"forbidden"} 403
curl -s -w ' %{{http_code}}\n' localhost:8080/api/users/5
# → {"error":"forbidden"} 403 (another student's
curl -s -w ' %{{http_code}}\n' localhost:8080/api/classes/1
# → exactly one name: "Aarav Sharma" (own row only)
curl -s -w ' %{{http_code}}\n' -X POST localhost:8080/api/c
-H 'Content-Type: application/json' -d '{"body":"hi"}'
# → {"error":"only the class teacher can do that"} 403

```

IF NOT: scoping middleware in `internal/handlers/scope.go`.

2.3 Teacher, read-only (T) — sign in as Ananya Iyer, `teacher1@gradetrack.dev`

T1 — Dashboard. EXPECT: "Your classes" with exactly **one** card, **Algebra II** (Mathematics), body "Ananya Iyer", "2026-27", **4 students** (active only — Ian excluded). IF NOT: `/classes` teacher filter or `student_active` on enrollments.

T2 — Class banner & tabs. Steps: click the card. EXPECT: banner "Algebra II — Mathematics · 2026-27 · Ananya Iyer · 4 students"; tabs **Stream | Performance | Marks | Attendance | Roster**; Stream opens first with a compose card "Announce something to your class" (textarea, counter "0 / 2000", **Post** disabled while empty) and the two seeded posts, each with a **Delete** button.

T3 — Performance tab. EXPECT, top card "Class overview":

- Horizontal bar chart "Average score by student", sorted: **Kabir Nair 95, Diya Patel 83.75, Aarav Sharma 78.4, Fiona Chen 37** (Fiona's bar **red**, others blue); dashed red line at 40%; x-axis labelled "Average score (%)". Ian is **not** in the chart.
- Scatter "Attendance vs average": 4 dots at (80, 78.4), (100,

83.75), (80, 95), (30, 37) — the last one red; dashed lines at $y=40$ and $x=75$. Hover a dot → name tooltip. "Trends" card: 4 sparklines with averages **95 / 83.75 / 78.4 / 37**, Fiona's line red. "Terms" card: grouped bars — Term 1 overall **72.5**, Term 2 overall **75**, plus category bars⁶. "Students" card: header **1 student at risk** (red); Export CSV button; rows ordered by name: Aarav 78.4 / 5 marks / 0 missing / 80% / 8 / 10 / OK; Diya 83.75 / 4 / **1** (red chip) / 100% / OK; **Fiona 37 / 4 / 1 / 30% / 3 / 10 / red "At risk" badge, row tinted red; Ian Brooks (inactive) 63.67 / 3 / 0 / 80% / 4 / 5 / grey "inactive" chip; Kabir 95 / 5 / 0 / 80% / OK. IF NOT: / performance, isAtRisk (avg < 40 or attendance < 75), PerformanceBar/Scatter.**

T4 — Performance export. Steps: click **Export CSV**. EXPECT: download `Algebra II performance.csv`; open it: header `Student, Status, Average %, Marks, Missing, Attendance %, Present, Sessions, At risk`, 5 rows, Fiona yes, Ian inactive. IF NOT: `lib/csv.js` / browser download blocked.

T5 — Marks tab (read parts). EXPECT three cards: "Enter scores" (assessment picker with 5 options like `Quiz 2 · quiz · Term 2 · /10`, "+ New assessment" link), "Score distribution" (picker defaults to **Quiz 2 · Term 2**; histogram buckets 0–39: 0, **40–59: 1, 60–74: 1, 75–89: 0, 90–100: 2**; dashed "mean 75%" marker; subtitle "4 scored · 0 missing · class mean 75%"⁷), and "All marks" with subtitle **23 shown · 2 missing** (20 active + 3 for Ian; Diya's Quiz 1 and Fiona's Assignment 2 are "missing" red chips). Newest rows first (2026-08-21). Ian's rows show **Ian Brooks (inactive)**. Filters: Term "Term 1" → 15 shown; Category "exam" → 5; both → 5; Term 2 + exam → empty state. Export CSV → `Algebra II marks Term 2 exam.csv` name reflects active filters. IF NOT: `/marks, Histogram, scoreDistribution`.

T6 — Attendance tab (read parts). EXPECT: "Attendance trend" line chart with 10 points: Aug 17 **80**, 18 **60**, 19 **80**, 20 **60**, 21 **80**, 24 **100**, 25 **50**, 26 **75**, 27 **75**, 28 **75**⁸ (Aug 17–21 include Ian's rows — see Known issue 5). "Take attendance" card with date defaulting to today, 4 checkboxes (no Ian), all ticked, "4 of 4 present". "Monthly register": month picker shows **August 2026**; grid columns 17 18 19 20 21 24 25 26 27 28; rows Aarav ✓✓✓x✓✓✓x✓✓ **8 / 10 80%**, Diya all ✓ **10 / 10 100%**, Fiona **xx✓xx✓x✓xx 3 / 10 30%**, Ian (inactive, greyed)

✓✓x✓✓ then ——— 4 / 5 80%, Kabir ✓x✓✓✓x✓✓✓ 8 / 10 80%;
footer "Present / total" 4/5 3/5 4/5 3/5 4/5 4/4 2/4 3/4 3/4 3/4. "Recorded
sessions — 10 dates" table newest first (2026-08-28: 3 present, 1
absent, 75%). Export CSV → Algebra II register 2026-08.csv.
IF NOT: `monthlyRegister / attendanceByDate` in `lib/stats.js`.

T7 — Roster tab (teacher = read-only). EXPECT: "Students — 4
active · 1 inactive"; Find box; **Show inactive** switch; 4 rows with green
`active` chips; no Enrol card, no Remove buttons. Tick Show inactive →
Ian Brooks (inactive) row appears greyed with an `inactive` chip.
Type `fi` in Find → only Fiona Chen. IF NOT: `RosterTab` role branch.

T8 — Forbidden surfaces. Steps: type `/classes/2` (Physics, owned
by Rohan). EXPECT: **forbidden** banner + back link. `/admin` →
redirected to `/teacher`. Curl:

```
T1=$(curl -s localhost:8080/api/auth/login -H 'Content-Type: application/json' -d '{"email": "teacher1@gradetrack.dev", "password": "password"}')
curl -s -w ' %{http_code}\n' localhost:8080/api/classes/2
# → {"error": "forbidden"} 403
curl -s -w ' %{http_code}\n' -X POST localhost:8080/api/assessment -H 'Content-Type: application/json' -d '{"student_id": 1, "score": 5}'
# → {"error": "only the class teacher can change assessment"} 403
curl -s -w ' %{http_code}\n' localhost:8080/api/users -H 'Content-Type: application/json' -d '{"email": "teacher1@gradetrack.dev", "password": "password"}'
curl -s -w ' %{http_code}\n' localhost:8080/api/admin/oversee -H 'Content-Type: application/json' -d '{"class_id": 2, "user_id": 1}'
curl -s -w ' %{http_code}\n' -X DELETE localhost:8080/api/announcement -H 'Content-Type: application/json' -d '{"class_id": 2, "text": "test"}'
# → {"error": "only the class teacher can delete announcement"} 403
```

IF NOT: `RequireClassOwner / RequireAssessmentOwner / RequireAnnouncementOwner`.

2.4 Teacher, mutations (TM) △ — still Ananya, Algebra II

Do these in order; later expectations depend on them.

TM1 — Bulk score entry with a new assessment. △ Steps: Marks →
"Enter scores" → click + **New assessment**. Fill title `Quiz 3`, category
`quiz`, term `Term 2`, Out of 10. The roster list appears: **Aarav, Diya,**
Fiona, Kabir (4, no Ian), "0 of 4 scored · out of 10". Type 11 for Aarav

→ inline **> 10** and the Save button disables with the message "Fix the highlighted scores first...". Change to 9. Diya 8, Kabir 10, leave Fiona blank. Click **Save all 4 scores**. EXPECT: toast **Saved 3 scores (1 missing)**; the picker now shows Quiz 3 · quiz · Term 2 · /10 selected with the values prefilled; "All marks" subtitle **27 shown · 3 missing**; four new rows dated today, Fiona's a red "missing" chip. Performance tab: Aarav **80.33**, Diya **83**, Kabir **95.83**, Fiona **37** (5 marks, 2 missing)⁹; still "1 student at risk". Score distribution picker now lists Quiz 3 (buckets 60–74: 0, 75–89: 1, 90–100: 2, "3 scored · 1 missing · class mean 90%"). IF NOT: POST /assessments/:id/scores (transaction) — if the request 400s, nothing at all is saved.

TM2 — Edit a single score. △ Steps: in "All marks", Aarav's Quiz 3 row → **Edit** → set 12 → inline **Must be 0–10**, Save disabled; set 7 → **Save**. EXPECT: toast **Score updated**; row shows 7 / 10, 70%. Performance: Aarav **77**⁹. IF NOT: PUT /marks/:id/RequireMarkOwner.

TM3 — Delete a single score. △ Steps: Diya's Quiz 3 row → **Delete** → confirm **Delete**. EXPECT: toast **Mark deleted**; subtitle **26 shown · 3 missing**; Diya back to **83.75** (4 marks). IF NOT: DELETE /marks/:id.

TM4 — Take attendance for a past date. △ Steps: Attendance → "Take attendance" → set date to 2026-09-02 (tomorrow or any future day). EXPECT: inline **Attendance cannot be recorded for a future date**, Save disabled. Set 2026-08-31, untick **Fiona Chen** ("3 of 4 present"), click **Save attendance**. EXPECT: toast **Saved attendance for 2026-08-31 — 3 of 4 present**; trend chart gains an **Aug 31 = 75%** point; register gains column **31** (Aarav ✓, Diya ✓, Fiona ✗, Kabir ✓, Ian —; footer 3/4); Recorded sessions now **11 dates**. Performance attendance: Aarav **81.82%** (9/11), Diya 100% (11/11), Fiona **27.27%** (3/11), Kabir ****81.82%****¹⁰. Re-select 2026-08-31 in the date picker → subtitle "This date is already recorded — saving will overwrite it" and Fiona's box is unticked (preloaded). IF NOT: POST /classes/:id/attendance upsert / future-date check.

TM5 — Post an announcement. △ Steps: Stream → type Test post from the guide → **Post**. EXPECT: toast **Announcement posted**; it appears at the top as "Ananya Iyer · just now"; Stream subtitle "3 announcements". Leave it for X1; it is deleted in D1. IF NOT: POST /classes/:id/announcements.

2.5 Cross-role integrity (X) — in the incognito window

X1 — Student sees the teacher's changes. Steps: sign in as Kabir (`student3@...`). EXPECT: Algebra II card **95.83%** / **81.82%**; block shows 6 marks, delta chip **↑ 5% vs Term 1**¹¹; heatmap has an 11th green cell **31** and **4 session streak**. Open class → Stream shows "Test post from the guide" by Ananya Iyer at the top. Marks tab: 6 rows, top one Quiz 3 10 / 10. IF NOT: caching in the browser (hard-reload) — otherwise API.

X2 — Missing mark is visible as missing. Steps: sign in as Fiona (`student4@...`). EXPECT: Algebra II **37%** (5 marks · 2 missing), attendance **27.27%**, red "At risk" badge, both big numbers red; Marks tab shows Quiz 3 as a red **missing** chip with "—" percent. IF NOT: nullable-score handling.

X3 — Aarav's edited score. Steps: sign in as Aarav (`student1@...`). EXPECT: Algebra II **77%** / **81.82%**; Marks: Quiz 3 7 / 10 70%; delta chip **↓ 7.3% vs Term 1** (Term 2 now 73.33 vs 80.67)¹¹.

2.6 Admin, read-only (AD) — sign in as Priya Menon, `admin@gradetrack.dev` (normal window)

AD1 — Overview stat cards. EXPECT (after TM1–TM4): Students **7 + the accounts you created in A6** (9 if you ran both the form and the curl), sub-line "N active" = that number minus 1 (lan); Teachers **2**; Classes **2**; School average ****74.25%****¹²; Attendance ****73.53%****¹²; **At risk 2** in red. (On a fresh seed, before §2.4: 74.33% and 73.44%.) IF NOT: `GET /admin/overview`.

AD2 — Overview charts & cards. EXPECT: "Classes at a glance" horizontal grouped bars, sorted by average: **Physics 101 75.11 / 75**, then **Algebra II 73.4 / 72.73**¹² (seed: 73.54 / 72.5). "Teacher load": Ananya Iyer 1 class / 4 students, Rohan Deshmukh 1 / 4. Below, "All classes" grid of 2 cards, each with the teacher's name and student count (4). IF NOT: `AdminCharts`, `teacherLoad`.

AD3 — Manage tab, Classes. Steps: click **Manage**. EXPECT: "Classes" card with a create form (Class name, Subject, Academic year

prefilled 2026-27, Teacher picker listing only Ananya Iyer and Rohan Deshmukh; **Create class** disabled until filled) and a table: 1 Algebra II Mathematics 2026-27 [Ananya ▼] 4; 2 Physics 101 Physics 2026-27 [Rohan ▼] 4.

AD4 — Users card, search & filters. EXPECT: "Users — N of N shown" where N = 10 seeded + accounts created in A6 (12 if you ran both); columns ID Name Email Roll no Role Active Created; Ian's row greyed "Ian Brooks (inactive)" with the switch off; the admin's own switch is **disabled**. Search hana → 1 row (server-side ?q=); search R100 → 7 rows (R1001–R1007); Role = Teacher → 2; Status = Inactive (search cleared) → Ian only. IF NOT: GET /users?q=.

AD5 — Admin on a class page = read-only. Steps: Overview → click Algebra II card. EXPECT: banner ends "· Ananya Iyer · 4 students"; all five tabs; Stream has **no compose box and no Delete**; Marks has **no "Enter scores"** card (Score distribution + All marks with Export only, no Edit/Delete); Attendance has **no "Take attendance"** card (trend + register + sessions only); Roster **has** an "Enrol a student" card and Remove buttons. Curl the documented restriction:

```
AD=$(curl -s localhost:8080/api/auth/login -H 'Content-Type: application/json' -d '{"email": "admin@gradetrack.dev", "password": "password"}')
curl -s -w ' %{http_code}\n' -X POST localhost:8080/api/courses -H 'Content-Type: application/json' -d '{"title": "x", "date": "2026-01-01"}'
# → {"error": "only the class teacher can do that"} 403
curl -s -w ' %{http_code}\n' -X POST localhost:8080/api/courses -H 'Content-Type: application/json' -d '{"date": "2026-01-01"}'
# → 403
```

IF NOT: RequireClassOwner (admin is deliberately excluded from grading).

2.7 Admin, mutations (ADM) ⚠

ADM1 — Create a class. ⚠ Steps: Manage → Class name Chemistry 101, Subject Chemistry, Teacher **Rohan Deshmukh** → **Create class**. EXPECT: toast **Created Chemistry 101**; table row 3 with **0** students;

Overview: Classes **3**, Teacher load Rohan **2 classes / 4 students**, a third (purple) card. Form resets. IF NOT: `POST /classes`.

ADM2 — Enrol a student. △ Steps: click the Chemistry 101 card → Roster → "Enrol a student" picker (lists active students not yet enrolled, with roll numbers) → **Aarav Sharma (R1001)** → **Enrol**. EXPECT: toast **Enrolled Aarav Sharma**; table shows him; banner "... · 1 student". Then enrol **Kabir Nair (R1003)** too (needed for E5). Picker no longer offers Ian (inactive). IF NOT: `POST /classes/:id/enrollments`.

ADM3 — Create a teacher with a temporary password. △ Steps: Manage → "Create a user": Full name Meera Krishnan, Email `meera@gradetrack.dev`, Role Teacher → **Create user**. EXPECT: green banner "Account for Meera Krishnan created. Temporary password: xxxxxxxxxxxx" (12 chars) and toast; Users table gains her row with a **temp password** chip. Copy the password. In incognito: sign in as `meera@gradetrack.dev` with it → lands on `/account/password` headed **Set a new password** with the blue notice; typing `/teacher` in the address bar bounces straight back here. Temporary password → new `MeeraPass2026` twice → **Save new password** → toast **Password changed** → `/teacher` with "No classes assigned yet". Log out. Back as admin (reload): the temp chip is gone. IF NOT: `POST /users, must_change_password` flow in `App.jsx` `RequireAuth`.

ADM4 — Reset a student's password. △ Steps: Users → Hana Kim → **Reset password**. EXPECT: blue banner with a new 12-char temporary password. Incognito: sign in as `student6@...` with `password123` → **invalid email or password** (old one dead); with the temp password → forced change screen; set new password `password123` (allowed: it only has to differ from the *temp* one) → `/student`. IF NOT: `PUT /users/:id/password`.

ADM5 — Deactivate / reactivate. △ Steps: Users → Ishaan Reddy → flip the **Active** switch off. EXPECT: toast **Deactivated Ishaan Reddy**; row greys with "(inactive)". Incognito: sign in as `student5@...` → **account deactivated**. Admin → Overview: the "active" sub-line drops by one; Physics 101 card **3 students**; Teacher load Rohan 2 classes / **5 students** (Physics 3 + Chemistry 2); "Classes at a glance": Physics **68.75 / 72.22**¹³; School average ****71.4%****¹³, Attendance ****72.58%****¹³, At risk still **2**. Open Physics 101 → Performance: Ishaan's row

"(inactive)" with grey chip, not in charts; header "2 students at risk" (Fiona, Hana). Roster: "3 active · 1 inactive". Flip the switch back on → toast **Reactivated Ishaan Reddy**; all numbers return. IF NOT: `PUT /users/:id/active; is_active` joins in `overview.go`.

ADM6 — CSV import. △ Steps: create `students.csv`:

```
name,email,roll_no
Neha Kapoor,neha@gradetrack.dev,R1008
Duplicate Person,student1@gradetrack.dev,R1009
,not-an-email,
```

Manage → "Import students from CSV" → choose the file. EXPECT: a 3-row preview (row 3 shows red "missing" chips for name). Click **Import 3 rows**. EXPECT: green banner "1 created. Shared temporary password: ..."; result table: row 1 **created**, row 2 **duplicate email**, row 3 **invalid**; toast "Imported 1 student"; Users search `neha` → 1 row with a temp-password chip. (Sign-in for Neha works like ADM3.) IF NOT: `POST /users/import, lib/csv.js` parser.

ADM7 — Reassign a class teacher. △ Steps: Manage → Classes table → Chemistry 101's Teacher dropdown → **Meera Krishnan**. EXPECT: toast **Reassigned Chemistry 101**; Teacher load now shows Meera 1 / 2, Rohan 1 / 4. (Only admins can do this; `only admins can reassign a class` is the API's answer to a teacher.)

2.8 Second cross-role pass (X2)

X4 — Teacher sees the new class. Sign in as Meera (`meera@... / MeeraPass2026`). EXPECT: one card Chemistry 101, "2 students". Its class page is used for the empty-state tests in §5.1.

X5 — Student sees the new enrolment. Sign in as Aarav. EXPECT: two cards (Algebra II, Chemistry 101); Chemistry card has no stats yet; "Across your classes" now has two groups (Chemistry avg "—").

3. Data-correctness spot checks

Run on a **fresh seed** (reset first). The same number must appear in every listed place.

Student / class	Metric	Expected	Where it must appear identically
Kabir Nair, Algebra II	average	95%	student card (S1) · student block big number (S3) · teacher Students table (T3) · bar chart tooltip (T3) · Trends row (T3)
Kabir Nair, Algebra II	attendance	80% (8/10)	S1 card · S3 "8 of 10 sessions" · T3 table · scatter x=80 · register "8 / 10 80%" (T6)
Fiona Chen, Algebra II	average	37%	S (as Fiona) big red number · T3 table (red row, At risk) · bar chart red bar
Fiona Chen, Algebra II	attendance	30% (3/10)	Fiona's heatmap "3 of 10" · T3 · register
Fiona Chen, Physics 101	average / attendance	58% / 66.67%	Fiona's Physics block · Rohan's Performance table (At risk — attendance < 75)
Algebra II, Term 1 filter	Fiona average	36%	curl GET /api/classes/1/performance?term=Term%201 as teacher1 (below)
Algebra II, Term 1	class overall (Terms chart)	72.5	T3 "Terms" card, Term 1 group, blue "Overall" bar

School	average / attendance / at risk	74.33% / 73.44% / 2	AD1 (fresh seed)
--------	--------------------------------	---------------------	------------------

Arithmetic: ^{2 6 12} in the footnotes. Term filter curl:

```
curl -s "localhost:8080/api/classes/1/performance?term=Te
# → ..."average_pct":36          (40 + 30 + 38) / 3
```

4. Security checks (curl)

Run from Git Bash. `$T1` is the teacher token from T8.

SEC1 — No / garbage token.

```
curl -s -w ' ${http_code}\n' localhost:8080/api/me
curl -s -w ' ${http_code}\n' localhost:8080/api/me -H 'Au
curl -s -w ' ${http_code}\n' localhost:8080/api/me -H "Au
```

Expired tokens: lifetime is 8 h — keep a token from today and retry tomorrow → 401. (Forging `alg=none` / `wrong-secret` tokens needs a script; both were verified during development.)

SEC2 — Token deleted mid-session (browser). Steps: signed in as any role, open DevTools → Application → Local Storage → delete token only → click any tab or card that fetches data. EXPECT: the next API call gets 401 and the app bounces you to `/login` (the `api.js` wrapper clears storage on 401).

SEC3 — Role-forbidden URLs. Covered by S9/S10, T8, AD5. Also:

```
curl -s -w ' ${http_code}\n' -X DELETE localhost:8080/api
curl -s -w ' ${http_code}\n' -X PUT localhost:8080/api/us
-H 'Content-Type: application/json' -d '{"is_active":fa
```

SEC4 — Login rate limit (5 per minute per IP + email). Uses a non-existent email so no account counter is touched:

```
for i in 1 2 3 4 5 6; do curl -s -o /dev/null -w '%{http_code}' \
-H 'Content-Type: application/json' -d '{"email":"nobody@example.com"}' \
# → 401 401 401 401 401 429          6th body: {"error":"too many requests"}
# wait 61 s, repeat once → 401 again
```

Note the limiter counts *all* attempts for that email, including successful ones (Known issue 9).

SEC5 — Account lockout (10 consecutive failures → 15 min). Uses `lan` (deactivated) so nothing else is affected. Two batches because of SEC4's window:

```
for i in 1 2 3 4 5; do curl -s -o /dev/null -w '%{http_code}' \
-H 'Content-Type: application/json' -d '{"email":"student7@gradetrack.dev"}' \
sleep 61
for i in 1 2 3 4 5; do curl -s -o /dev/null -w '%{http_code}' \
-H 'Content-Type: application/json' -d '{"email":"student7@gradetrack.dev"}' \
sleep 61
curl -s -w '%{http_code}\n' localhost:8080/api/auth/login \
-d '{"email":"student7@gradetrack.dev","password":"password"}' \
# → {"error":"account temporarily locked"} 423 ← even w/ correct password
```

Clear it: `mysql.exe -u root -p gradetrack -e "UPDATE users SET locked_until=NULL, failed_attempts=0 WHERE id=10"` (or reset the DB). After clearing, the correct password gives **403 account deactivated** again.

SEC6 — Oversized body (1 MB cap).

```
head -c 2100000 /dev/zero | tr '\0' 'x' > big.txt
curl -s -w '%{http_code}\n' -X POST localhost:8080/api/courses \
-H 'Content-Type: application/json' --data-binary @big.txt \
# → {"error":"request too large"} 413
rm big.txt
```

SEC7 — Security headers & CORS.

```
curl -s -i localhost:8080/api/health | grep -iE '^(x-cont
# → nosniff / DENY / no-referrer / a default-src 'self' ...
curl -s -w ' %{{http_code}}\n' -X OPTIONS localhost:8080/ap
# → {"error":"origin not allowed"} 403
curl -s -o /dev/null -w ' %{{http_code}}\n' -X OPTIONS local
# → 204
```

5. Edge & empty states

5.1 Empty states (E) — Chemistry 101 exists from ADM1/ADM2 with Aarav + Kabir enrolled, no data

E1 — Empty class, teacher view. Sign in as Meera (owner after ADM7), open Chemistry 101. EXPECT: Stream → **Nothing posted yet** ("Your first announcement will show up here."). Performance → both charts show **Not enough data yet**; Trends rows read "no marks" with "—"; Terms → **Needs 2+ terms to compare**; Students table: Aarav and Kabir with "—" averages, "0 / 0", status OK; header **No students at risk** (green). Marks → Score distribution **No assessments yet**; All marks **No marks yet** ("Enter scores above."). Attendance → trend **Not enough data yet**; register **No sessions recorded**; sessions **No attendance recorded yet**.

E2 — One term only → term comparison still empty. △ As Meera: Enter scores → + New assessment *Intro Quiz*, *quiz*, *Term 1*, out of 10; Aarav 6, Kabir 9 → Save all 2 scores. EXPECT: Terms card still **Needs 2+ terms to compare** (one term); Trends rows now say "1 mark" (a sparkline needs 2); distribution shows *Intro Quiz*, "2 scored · 0 missing · class mean 75%".

E3 — Sparkline with one point, student view. Sign in as Aarav → Chemistry 101 block: average **60%**, "Marks over time" has a single dot, "Attendance calendar" **Not enough data yet**, streak 0; Term

comparison **Needs 2+ terms to compare**; category bars: quizzes 60 only.

E4 — Student with no classes. Sign in as `teststudent@gradetrack.dev / testpass123 (A6)`. EXPECT: **Not enrolled in any classes yet** and nothing else; `/classes/1` → **forbidden**.

E5 — Radar at three classes. Sign in as Kabir (now Algebra II + Physics 101 + Chemistry 101 with a 90% Intro Quiz). EXPECT: "Across your classes" renders a **radar** (triangle) with axes Algebra II 95.83, Physics 101 84, Chemistry 101 90 — no attendance series in radar mode.

E6 — Teacher with no classes. Sign in as Rohan after ADM7 gave Chemistry away: still 1 card (Physics). A teacher with zero classes (e.g. a freshly created one) sees **No classes assigned yet**.

E7 — Filter no-match. Any Marks tab: Term 2 + exam → **No marks match these filters**.

E8 — Roster / user searches with no hits. Roster Find `zzz` → **No students match**; Users search `zzz` → **No users match**.

5.2 Form validation (V) — rules as implemented; one bad input each

Form	Rule (source)	Try
Login	both fields required; server <code>required,max=255/72</code>	leave password empty
Register	email format (client, on blur)	<code>foo@bar</code> → tab away

Register	password ≥ 8 (client + server min=8, max=72)	abc
----------	---	-----

Register	roll_no [A-Za-z0-9-]{1,30} (server)	R 1!
Register	name ≤ 100 , email ≤ 255 (server)	101-char name
Change password	new ≥ 8 / differs / confirm matches	see S8
Create class	name, subject, teacher required; year ≤ 9	leave Teacher unset
Create user	email format / duplicate	student1@gradetrack.dev
Create user	roll_no only for students (server)	curl { "role": "teacher", "roll_
Enter scores (bulk)	0 $\leq$ score $\leq$ max (client)	-1 / 11 on a /10

Enter scores (new assessment)	max > 0	Out of 0

Enter scores (new assessment)	category ∈ assignment/ quiz/exam (server)	curl "category": "lab"
Edit mark	$0 \leq \text{score} \leq \text{max}$	12 on /10
Take attendance	date ≤ today (client + server)	tomorrow
Announcement	non-empty, ≤ 2000 chars	empty

Import CSV	name & email present, email has @, unique email/roll	rows in ADM6
Enrol	student must be active (server)	curl {"student_id":10}

Curl forms for the server-only rows:

```
curl -s localhost:8080/api/users -H "Authorization: Beare
-d '{"name":"X","email":"x1@gradetrack.dev","role":"tea
curl -s localhost:8080/api/classes/1/assessments -H "Auth
-d '{"title":"x","category":"lab","term":"Term 1","max_
curl -s localhost:8080/api/classes/1/enrollments -H "Auth
-d '{"student_id":10}'
curl -s -X PUT localhost:8080/api/users/1/active -H "Auth
-d '{"is_active":false}'
```

5.3 Destructive paths — do these LAST (D) ⚠

D1 — Delete an announcement. As Ananya, Algebra II → Stream → "Test post from the guide" → **Delete** → confirm. EXPECT: toast **Announcement deleted**; back to 2 announcements.

D2 — Remove an enrolment. As admin → Chemistry 101 → Roster → Aarav → **Remove** → confirm. EXPECT: toast **Removed Aarav Sharma**; his Intro Quiz mark stays in the DB but he no longer sees the class (Aarav's dashboard: 1 card).

D3 — Edit / delete an assessment (API only — no UI).

```
M=$(curl -s localhost:8080/api/auth/login -H 'Content-Typ
-d '{"email":"meera@gradetrack.dev","password":"MeeraPa
AID=$(curl -s localhost:8080/api/classes/3/assessments -H
```

```
curl -s -w ' ${http_code}\n' -X PUT localhost:8080/api/as
  -d '{"title":"Intro Quiz","category":"quiz","term":"Ter
# → {"error":"existing scores exceed the new max_score"}
curl -s -w ' ${http_code}\n' -X PUT localhost:8080/api/as
  -d '{"title":"Intro Quiz (renamed)","category":"quiz","
curl -s -w ' ${http_code}\n' -X POST localhost:8080/api/a
  -d '{"student_id":6,"score":null}'
curl -s -o /dev/null -w '${http_code}\n' -X DELETE localh
```

EXPECT: Meera's Marks tab → "No assessments yet" again; every mark on it is gone (cascade), and each deletion was written to `mark_audit` (write-only: `SELECT * FROM mark_audit` shows rows with `new_score NULL`).

D4 — Delete a class (API only — no UI).

```
curl -s -o /dev/null -w '${http_code}\n' -X DELETE localh
curl -s -w ' ${http_code}\n' localhost:8080/api/classes/3
```

EXPECT: admin Overview → Classes **2**; Meera's dashboard → **No classes assigned yet**.

D5 — Reset. Run the two `mysql.exe` lines from §1.1. Confirm with A3 + T3 (Aarav back to 78.4).

6. Coverage checklist

API endpoints (`internal/handlers/routes.go`)

Method	Path	Tests
GET	/api/health	§1.1, SEC7
POST	/api/auth/register	A6, A7, V
POST	/api/auth/login	A2–A4, SEC4, SEC5

GET	/api/me	SEC1 (curl; no UI uses it)
PUT	/api/me/password	S8, ADM3, ADM4
GET	/api/admin/overview	AD1, AD2, ADM5, T8 (403)
GET	/api/users (+?q=)	AD4, S10 (403), T8 (403)
POST	/api/users	ADM3, V
POST	/api/users/import	ADM6
GET	/api/users/:id	S10 (self 200 / other 403); admin class banner AD5
PUT	/api/users/:id/ password	ADM4
PUT	/api/users/:id/active	ADM5, SEC3, V (self)
POST	/api/classes	ADM1, AD3
GET	/api/classes	S1, T1, AD2, X4, X5
GET	/api/classes/:id	S4, S9, T8, D4
PUT	/api/classes/:id	ADM7
DELETE	/api/classes/:id	D4, SEC3 (403)
POST	/api/classes/:id/ enrollments	ADM2, V (inactive)
GET	/api/classes/:id/ enrollments	T7, AD5, S10 (403)
DELETE	/api/enrollments/:id	D2
POST	/api/classes/:id/ assessments	TM1, E2, AD5 (403), V
GET	/api/classes/:id/ assessments	T5, D3
PUT	/api/assessments/:id	D3
DELETE	/api/assessments/:id	D3
POST	/api/assessments/:id/ marks	D3 (single score / missing)

POST	/api/assessments/:id/scores	TM1, E2, T8 (403)
GET	/api/classes/:id/marks	S6, T5, S10 (403), T8 (403)
PUT	/api/marks/:id	TM2
DELETE	/api/marks/:id	TM3
POST	/api/classes/:id/attendance	TM4, AD5 (403), V
GET	/api/classes/:id/attendance	S7, T6
GET	/api/classes/:id/performance	S1, S3, T3, §3 (?term=), S10
POST	/api/classes/:id/announcements	TM5, S10 (403), SEC6
GET	/api/classes/:id/announcements	S4, T2, X1
DELETE	/api/announcements/:id	D1, T8 (403)
—	NoRoute → 404 JSON	curl -s localhost:8080/api/nope → {"error": "not found"}

Pages (web/src/App.jsx)

Route	Component	Tests
/	Home redirect	A1, A3
/login	Login	A1–A4
/register	Register	A6, A7, V
/account/password	ChangePassword	S8, ADM3, ADM4, A5

/student	StudentDashboard	S1–S3, X1–X3, E3–E5
/teacher	TeacherDashboard	T1, X4, E6
/admin	AdminDashboard (Overview + Manage)	AD1–AD4, ADM1–ADM7
/classes/:id — Stream tab	StreamTab	S4, T2, TM5, D1, E1
/classes/:id — Performance tab	PerformanceTab / StudentSummary	S5, T3, T4, E1, E2
/classes/:id — Marks tab	MarksTab (bulk, distribution, table)	S6, T5, TM1–TM3, E2, E7
/classes/:id — Attendance tab	AttendanceTab (trend, take, register)	S7, T6, TM4, E1
/classes/:id — Roster tab	RosterTab	T7, AD5, ADM2, D2, E8
* (unknown)	Home redirect	A5
App bar / account menu	AppBar	A3, S8

7. Known issues found while writing this guide

Nothing here was fixed — listed as found.

1. **No UI for four endpoints:** DELETE /classes/:id, PUT /assessments/:id, DELETE /assessments/:id, POST /assessments/:id/marks (single score) and GET /me are API-only. Deleting a class or renaming an assessment requires curl (D3, D4).
2. **Teachers cannot enrol students.** The API allows the owning teacher to POST /classes/:id/enrollments, but the picker

needs `GET /users` (admin-only), so the Roster tab is read-only for teachers.

3. **Students never see a class average.** `/performance` returns only the student's own row, so the standing line reads "Class average: not shared with students" and category bars are personal-only.
 4. **Histogram mean marker snaps to a bucket** (category axis): the dashed "mean 75%" line sits at the centre of the 75–89 bucket, not at the exact value.
 5. **Inactive students' history leaks into class-level attendance %.** The trend chart and register include Ian's Aug 17–21 rows (5-student denominators), while the at-risk count, bar/scatter charts and the admin overview exclude inactive students. Numbers are individually correct but the two views use different populations.
 6. **User search passes % and _ to SQL LIKE unescaped** — typing % in the admin search matches every user. Harmless (admin-only, parameterised) but surprising.
 7. **Password reset / deactivation don't revoke live tokens.** An already-signed-in user keeps working for up to 8 h after an admin resets their password or deactivates them (the token is stateless); only the next login is blocked.
 8. **Overwriting a score keeps the original `created_at`,** so the "Recorded" date in the marks table doesn't change when a bulk entry updates an existing mark (the change is in `mark_audit` instead).
 9. **The login rate limiter counts successful logins too:** six sign-ins in one minute with the correct password returns 429 on the sixth. Documented behaviour, but it will bite anyone switching roles quickly in one browser.
 10. **Lock-before-active ordering:** a deactivated account can still be "temporarily locked" (SEC5); cosmetic, since it can't sign in either way.
 11. **Stream relative times use the browser clock** ("4 days ago" drifts as you test on later dates); the absolute time is in the hover title.
-

8. Footnotes — arithmetic

All percentages are $\text{score} / \text{max_score} \times 100$, averaged per student (NULL = missing, excluded from the average but counted as missing). Attendance % = present / recorded sessions.

¹ Assessment list is verbatim from `seed.sql` (`INSERT INTO assessments`).

² Seed averages (all terms):

- Aarav (Algebra): $18/20=90$, $8/10=80$, $72/100=72$, $16/20=80$, $7/10=70 \rightarrow 392/5 = \mathbf{78.4}$; attendance absent Aug 20 & 26 $\rightarrow 8/10 = \mathbf{80}$.
- Diya: $15/20=75$, Quiz 1 NULL, 85 , $17/20=85$, $9/10=90 \rightarrow 335/4 = \mathbf{83.75}$, 1 missing; $10/10 = \mathbf{100}$.
- Kabir (Algebra): $100, 90, 90, 95, 100 \rightarrow 475/5 = \mathbf{95}$; absent Aug 18 & 25 $\rightarrow \mathbf{80}$. Kabir (Physics): $45/50=90, 80, 78, 41/50=82, 90 \rightarrow 420/5 = \mathbf{84}$; $6/6 = \mathbf{100}$.
- Fiona (Algebra): $8/20=40, 3/10=30, 38$, Assignment 2 NULL, $4/10=40 \rightarrow 148/4 = \mathbf{37}$; present Aug 19, 24, 26 $\rightarrow 3/10 = \mathbf{30}$. Fiona (Physics): $30/50=60, 60, 62$, Lab 2 NULL, $50 \rightarrow 232/4 = \mathbf{58}$; absent Aug 19 & 26 $\rightarrow 4/6 = \mathbf{66.67}$.
- Ishaan (Physics): $96, 100, 91, 94, 90 \rightarrow 471/5 = \mathbf{94.2}$; absent Aug 27 $\rightarrow 5/6 = \mathbf{83.33}$.
- Hana (Physics): $76, \text{NULL}, 55, 66, 60 \rightarrow 257/4 = \mathbf{64.25}$; present Aug 19, 25, 27 $\rightarrow 3/6 = \mathbf{50}$.
- Ian (Algebra, inactive): $14/20=70, 6/10=60, 61 \rightarrow 191/3 = \mathbf{63.67}$; $4/5 = \mathbf{80}$. At risk = average < 40 **or** attendance < 75: Algebra $\rightarrow$ Fiona; Physics $\rightarrow$ Fiona (66.67) and Hana (50).

³ Term deltas (latest term minus previous, personal): Kabir Algebra Term 1 $(100+90+90)/3 = 93.33$, Term 2 $(95+100)/2 = 97.5 \rightarrow \mathbf{+4.2}$. Kabir Physics T1 $(90+80+78)/3 = 82.67$, T2 $(82+90)/2 = 86 \rightarrow \mathbf{+3.3}$.

⁴ Streak = consecutive "present" counting back from the latest session. Kabir Algebra: 28 ✓, 27 ✓, 26 ✓, 25 ✗ $\rightarrow \mathbf{3}$. Aarav: 28 ✓, 27 ✓, 26 ✗ $\rightarrow \mathbf{2}$. Diya 10. Fiona: 28 ✗ $\rightarrow \mathbf{0}$.

⁵ Kabir Term 1 categories: assignment 100, quiz 90, exam 90; Term 2: assignment 95, quiz 100 (no Term 2 exam).

⁶ Class-wide term averages (each student averaged first, then students

averaged; active students only): Term 1 — Aarav 80.67, Diya $(75+85)/2 = 80$, Kabir 93.33, Fiona 36 $\rightarrow 290/4 = \mathbf{72.5}$. Categories: assignments $(90+75+100+40)/4 = 76.3$; quizzes $(80+90+30)/3 = 66.7$ (Diya's quiz is missing); exams $(72+85+90+38)/4 = 71.3$. Term 2 — Aarav 75, Diya 87.5, Kabir 97.5, Fiona 40 $\rightarrow 300/4 = \mathbf{75}$; assignments $(80+85+95)/3 = 86.7$; quizzes $(70+90+100+40)/4 = 75$.

⁷ Quiz 2 (Algebra, /10), active students: Aarav 70, Diya 90, Kabir 100, Fiona 40 $\rightarrow$ buckets 40–59: 1, 60–74: 1, 90–100: 2; mean $(70+90+100+40)/4 = \mathbf{75}$.

⁸ Algebra attendance per date (all recorded rows incl. Ian for Aug 17–21): 17 $\rightarrow 4/5 = 80$; 18 $\rightarrow 3/5 = 60$; 19 $\rightarrow 4/5 = 80$; 20 $\rightarrow 3/5 = 60$; 21 $\rightarrow 4/5 = 80$; 24 $\rightarrow 4/4 = 100$; 25 $\rightarrow 2/4 = 50$; 26 $\rightarrow 3/4 = 75$; 27 $\rightarrow 3/4 = 75$; 28 $\rightarrow 3/4 = 75$.

⁹ After TM1 (Quiz 3 /10: Aarav 9, Diya 8, Kabir 10, Fiona missing): Aarav $(392+90)/6 = \mathbf{80.33}$; Diya $(335+80)/5 = \mathbf{83}$; Kabir $(475+100)/6 = \mathbf{95.83}$; Fiona unchanged 37 (2 missing). Quiz 3 distribution: 90, 80, 100 $\rightarrow$ mean 90. After TM2 (Aarav 9 $\rightarrow$ 7): $(392+70)/6 = \mathbf{77}$. After TM3 Diya returns to 83.75.

¹⁰ After TM4 (Aug 31: everyone present except Fiona): Aarav 9/11 = **81.82**, Diya 11/11 = 100, Kabir 9/11 = **81.82**, Fiona 3/11 = **27.27**. Class $3/4 = 75\%$ for the new date. Streaks: Kabir 4, Aarav 3, Diya 11.

¹¹ Deltas after TM1–TM3: Kabir Term 2 $(95+100+100)/3 = 98.33$ vs 93.33 $\rightarrow \mathbf{+5.0}$. Aarav Term 2 $(80+70+70)/3 = 73.33$ vs 80.67 $\rightarrow \mathbf{-7.3}$.

¹² School numbers (active enrolments only: 4 in Algebra, 4 in Physics). Fresh seed: averages 78.4, 83.75, 95, 37, 84, 58, 94.2, 64.25 $\rightarrow 594.6/8 = \mathbf{74.33}$; attendance present/sessions $(8+10+8+3)+(6+4+5+3) = 47$ of $40+24 = 64 \rightarrow \mathbf{73.44}$; at-risk distinct students: Fiona, Hana $\rightarrow \mathbf{2}$. Per class: Algebra $(78.4+83.75+95+37)/4 = \mathbf{73.54}$, $29/40 = \mathbf{72.5}$; Physics $(84+58+94.2+64.25)/4 = \mathbf{75.11}$, $18/24 = \mathbf{75}$. After TM1–TM4 (Aarav 77, Diya 83.75, Kabir 95.83, Fiona 37; Algebra attendance 32/44): school average $(77+83.75+95.83+37+84+58+94.2+64.25)/8 = 594.03/8 = \mathbf{74.25}$; Algebra class average $(77+83.75+95.83+37)/4 = 293.58/4 = \mathbf{73.4}$, attendance 32/44 = **72.73**; school attendance $(32+18)/(44+24) = 50/68 = \mathbf{73.53}$. (Verified by replaying TM1–TM4 through the API while writing this guide.)

¹³ With Ishaan deactivated (ADM5), on the post-TM1–TM4 data. Physics active = Kabir, Fiona, Hana: class average $(84+58+64.25)/3 = 206.25/3 = \mathbf{68.75}$, attendance $(6+4+3)/18 = \mathbf{72.22}$. School average over the 7 active enrolments $(77+83.75+95.83+37+84+58+64.25)/7 = 499.83/7 = \mathbf{71.4}$; school attendance $(32+13)/(44+18) = 45/62 = \mathbf{72.58}$. Everything returns to the ¹² values when he is reactivated.